



Find the Loop Bug

Quick facilitation notes & answer key

Goal

Students read a buggy loop, decide whether the bug is the repeat count or an inside block, and fix it so the code matches the goal (Ontario C3.2).

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project — show a buggy loop and fix it live; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Hook: read directions with a wrong step — finding the mistake is debugging.
2. Project the buggy repeat-3 (should be 5) loop; find and fix it together.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: say what it SHOULD do, then find the wrong part.

Answer key (checked by the run-gate)

Ex 1 — the repeat number is wrong; it should be repeat 5.

Ex 2 — the second block is wrong; change "stamp red" to "stamp blue".

Ex 3 — repeat 3 ($3 \times 10 = 30$).

Ex 4 (challenge) — the inside is the bug: two move blocks; remove one (repeat $4 \times 1 = 4$ moves).

Watch for & differentiate

Common slip: changing the repeat number in Ex 4 when the bug is the extra inside block.

Support: write the GOAL number first, then compare to what the code does.

Extend: a different Ex 4 fix — repeat 2 with two moves also gives 4.

Success looks like

The student says what the code should do, finds whether the bug is the count or an inside block, and fixes it to match the goal (Ex 1–4).